

PARVEZ AHMED

Software Engineer · Agentic Systems & Applied LLM · TLM Reconciliations

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SUMMARY

Builds agentic LLM systems inside production financial infrastructure. Shipped an agentic RAG layer over a legacy AS/400 codebase on a Neo4j knowledge graph, trained an ML reconciliation model that resolves break patterns the TLM matching engine cannot, and support reconciliation activities around Smartstream TLM. Runs two open-source document-intelligence and reconciliation services scored on held-out data behind CI accuracy gates. Backend depth — Python, Java, Spring Boot, SQL — from a global custodian bank.

PROFESSIONAL EXPERIENCE

Senior Analyst, Software Engineer

Apr 2026 – Present

Northern Trust · Bengaluru, Karnataka · Promoted from Analyst, Software Engineer

- **Built an agentic RAG system over a decades-old AS/400 codebase:** Azure OpenAI models, a Neo4j knowledge graph, LLM-generated Cypher retrieval and a custom chat interface that lets developers query legacy business logic in natural language.
- **Trained an ML matching model on historical manual-match data** that resolves the one-to-many and many-to-many break patterns the TLM auto-matching engine cannot. Clears over 21000 breaks/week at 99.62% match rate on production data, increasing STP by 1.28%.
- **Upgraded 16 Java microservices from Java 8 → 21 and Spring Boot 2.x → 3.x** with zero production downtime; remediated 800+ CVEs across dependencies and container configurations.

Analyst, Software Engineer

Aug 2024 – Apr 2026

Northern Trust · Bengaluru, Karnataka

- **Engineered an ISO 20022 → CAMT.053 conversion pipeline** parsing large financial XML into optimised chunks, cutting processing from ~9 minutes to under 50 seconds (91% reduction).
- **Built a Python report-conversion tool** turning large TLM HTML reports into optimised XLSX, cutting load time from minutes to under 5 seconds. Adopted company-wide by operations teams.
- **Led zero-downtime migration of 21 Spring Boot microservices (v2.1.x → v2.7.x)** and the RHEL7 → RHEL8 upgrade of production Unix servers carrying 30+ upstream dependencies.
- **Automated CI/CD via GitHub Actions** across PCF-hosted services, cutting deployment time 40%.

Software Engineer, Contract

Jun 2026 – Sep 2026

CaptainSide · Remote · Consumer game-discovery platform

- **Designed and shipped the Discover recommendation engine:** 19 TypeScript modules across a three-stage retrieval → ranking → rerank pipeline, with an EMA taste model, a satisfaction controller and a session-scoped ring state machine. Validated against a 14-check offline simulation harness, then rolled out behind a feature flag.
- **Found and fixed an unauthenticated account-takeover** in the password-reset route, then closed the same bug class across seven mutation routes by deriving identity from the session rather than the request body.

Project Intern

Jun 2023 – Sep 2023

Northern Trust · Bengaluru, Karnataka

- **Raised frontend Lighthouse scores 25%** through targeted audits, code-splitting and accessibility fixes; integrated REST APIs powering dynamic financial dashboards.

Project Collaborator

Oct 2022 – Mar 2023

Volkswagen Group New Academy · Chakan, Pune, Maharashtra

- **Built a full-stack Decision Support System (Django, React, PostgreSQL, Docker)** for sales and HR analytics across 100+ dealership branches, replacing manual CSV aggregation with real-time reporting.

PROJECTS

DocVal — Bank-Statement Extraction with Arithmetic Validation

Python · FastAPI · Docker

Live: docval-yy4s.onrender.com · Code: github.com/Sero01/docval

- **Converts PDF bank statements into typed JSON**, routing each page between native text-layer parsing and vision-LLM extraction, then validating that opening balance plus every transaction equals the closing balance.
- **Held-out F1 0.63 overall** — 0.92 on digital PDFs, 0.42 on dense scans — at ~\$0.008/document. 61 tests behind a CI gate that fails the build on accuracy regression.

ReconMatch — Transaction Reconciliation & Break Classification

Python · matching engine

Live: reconmatch-aa9c.onrender.com · Code: github.com/Sero01/reconmatch

- **Deterministic engine aligning bank-statement lines to ledger entries** through tiered rules (exact, date-windowed, split/partial, N:1 exact-sum batch), with a greedy assignment pass preventing double-allocation and typed classification of every unmatched item.

- **Resolved 88.65% of the BenchRec cash v1.0 held-out set** under strict exact-set scoring, against 62.45% for the published reference. 51 tests.

tiny-router — Can a 26M-Parameter Model Be a Safe First Stage?

Python · LLM evaluation

- **Tested a ~26M-parameter model as the cheap first stage of an LLM workflow**, emitting one line of a closed typed action DSL that a hand-written validator and tier policy turn into execute, draft or escalate.
- **1,028-example evaluation corpus** with a risk/coverage sweep and ablation ladder. Calibration drove expected calibration error to near zero and changed unsafe executions by exactly zero; treating message bodies as untrusted cut them from 11 to 7.

Emma & Council — Local-First Agent Tooling

Python · FastAPI · Ollama

- **Emma**: self-hosted agent on Gemma via Ollama with a plan-first PLAN → EXECUTE → SYNTH loop, six tools, FastAPI server and SQLite persistence. **Council**: three-way terminal workflow across two coding agents and one codebase. Both fully offline.

agentguides.dev — Weekly Publication on the AI Agent Ecosystem

Astro · Cloudflare Pages

- **Built the site and the cron-driven pipeline** that has published every Sunday for six months without a miss: LLM leaderboards, model reviews, AI coding-tool comparisons.

TECHNICAL SKILLS

AI / ML: LLMs, RAG, graph retrieval, agentic workflows, prompt engineering, ML model training, evaluation frameworks (held-out splits, calibration, risk/coverage), Azure OpenAI, Ollama

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash

Backend & Web: FastAPI, Spring Boot, Django, Node.js, REST APIs, microservices, React, Next.js

Data: PostgreSQL, MySQL, SQLite, Neo4j, Prisma, XML/JSON/CSV processing, XLSX automation, ISO 20022 / CAMT.053

Cloud & DevOps: Docker, GitHub Actions CI/CD, Git, Linux/RHEL, Azure, Pivotal Cloud Foundry, Cloudflare, pytest/JUnit, CVE remediation

Domain: Financial reconciliation (TLM), custodian banking operations, decision-support systems, enterprise integration

EDUCATION

B.Tech, Computer Science Engineering

Sep 2020 — May 2024

Dr. Vishwanath Karad MIT World Peace University · Pune, Maharashtra · CGPA 9.55 / 10 · Top-6 finalist, Northern Trust Intern Hackathon

CERTIFICATIONS

Claude Code in Action

2026

Anthropic

Deploying Claude Enterprise with Confidence

2026

Anthropic